

# Differential CC $1\mu 1\pi^\pm$ cross sections on argon in the NuMI beam

Status of the blind analysis and a request to open the control regions

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$\nu_\mu + \bar{\nu}_\mu$  CC single charged pion on argon • FHC, RHC and combined • blind, fake-data results

# What is measured

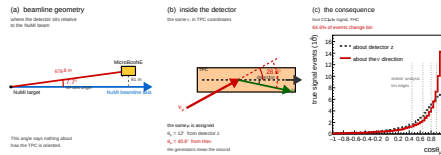
**Signal** (truth level, charge-blind):  $\nu_\mu/\bar{\nu}_\mu$  CC in the FV with exactly one  $\mu$ , exactly one  $\pi^\pm$ , no  $\pi^0$ , no kaons or other mesons, any number of nucleons; coherent production included.

**Measured phase space:**  $p_\mu > 0.15$ ,  $p_\pi > 0.175$  GeV/c,  $\theta_{\mu\pi} < 2.6$  rad

## Deliverables per configuration (FHC, RHC, combined)

- ▶ total flux-averaged cross section (one-bin extraction, no regularisation)
- ▶  $d\sigma/dp_\mu$ ,  $d\sigma/d \cos \theta_\mu$  and  $d\sigma/d\theta_\mu$ ,  $d\sigma/d \cos \theta_\pi$ ,  $d\sigma/d\theta_{\mu\pi}$ , two-bin  $d\sigma/dp_\pi$
- ▶ proton-tagged subsample: integrated cross section (headline); TKI and  $W_{\text{had}}$  as secondary results

**Blind analysis:** every “data” point in this talk is Poisson-thrown central-value MC. No beam-on signal region has been opened.



MicroBooNE sits  $7.7^\circ$  off the NuMI axis; the beam arrives  $28^\circ$  from detector  $z$ , so all angles and transverse quantities are computed in the beam frame.

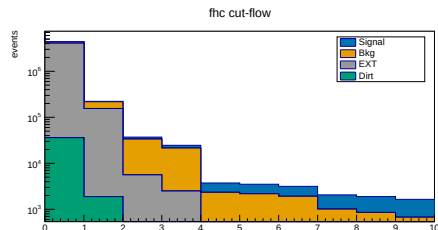
# Selection and yields

**Cuts:** software trigger, vertex in FV, topological score  $> 0.67$ , muon candidate (highest muon-BDT score among qualifying tracks), exactly one contained pion candidate (LLR, MIP-BDT, pion-BDT, Bragg- $\pi$ ), three-plane hits, shower veto,  $\theta_{\mu\pi} < 2.6$  rad, non-proton multiplicity  $< 4$  and  $< 5$  tracks.

| Final cut | Signal | $\nu$ bkg | EXT | Pred. | Pur.  |
|-----------|--------|-----------|-----|-------|-------|
| FHC       | 943    | 637       | 30  | 1 612 | 58.5% |
| RHC       | 1 015  | 720       | 23  | 1 759 | 57.7% |
| Comb.     | 1 959  | 1 357     | 53  | 3 372 | 58.1% |

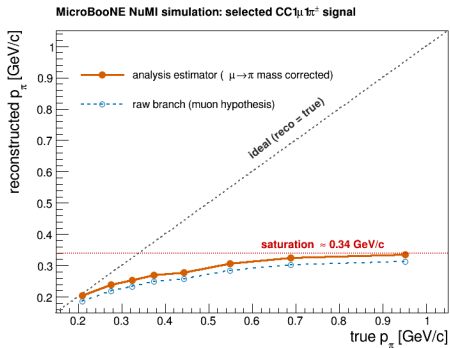
Efficiency 17.4–17.6% in the measured phase space; largest background: CC0 $\pi$  with a proton taken as the pion.

**Normalisation** (details in backup): 8.86 (FHC) and  $11.08 \times 10^{20}$  POT (RHC), each run scaled by its own POT; per-configuration new-G4 flux; beam-off matched by run period and scaled by gates (1.9% of the sample).

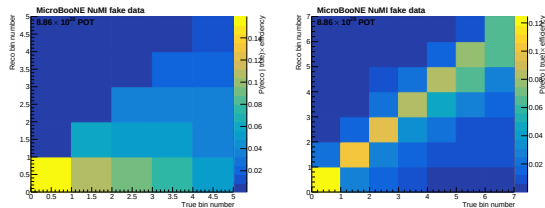


FHC cut-flow, per-run POT-scaled, full CV weight.

# The pion momentum sets the scope



Reconstructed  $p_\pi$  (range under the muon hypothesis, mass-corrected) saturates near 0.34 GeV/c: pions interact before stopping.



Smearacceptance  $S = \varepsilon M$ : five-bin  $p_\pi$  (diagnostic, left) vs.  $p_\mu$  (right).

- ▶ every estimator tried (golden-pion tag, MCS, calorimetry, regression, daughter recovery): a physics limit, not an algorithmic one
- ▶  $p_\pi$  released as **two bins**:  $[0.175, 0.205]$  and  $> 0.205$  GeV/c, migration diagonals 91% / 76%
- ▶ proton-tagged  $W_{\pi p}$  shape **withdrawn**; the sample supports an integrated cross section

# Extraction: Wiener-SVD and what its integral means

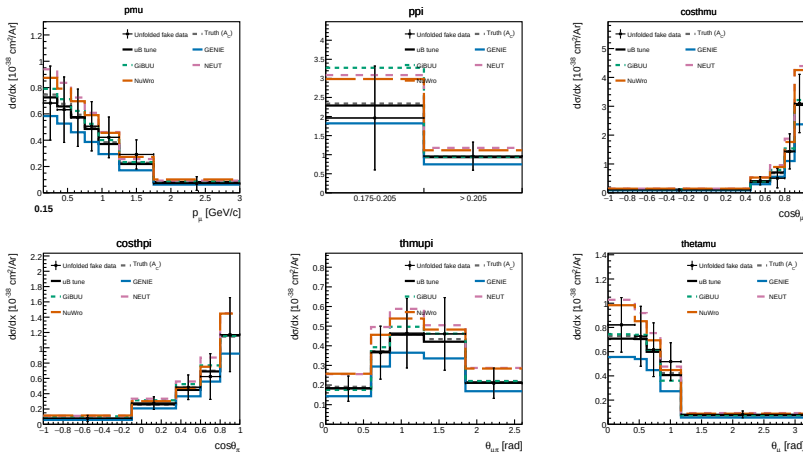
- ▶  $n^{\text{reco}} = S N^{\text{true}} + b$  with  $S = \varepsilon M$  inverted once (no extra  $1/\varepsilon$ )
- ▶ Wiener-SVD (XSECANALYZER), second-derivative regularisation aware of the non-uniform bin widths
- ▶ the result is  $A_C \times_{\text{true}}$ ; every prediction is smeared by the released  $A_C$  before comparison
- ▶  $A_C$  is *not* norm preserving: the integral of a differential result is observable-dependent ( $0.62\text{--}0.96 \times 10^{-38} \text{ cm}^2/\text{Ar}$ ); D'Agostini gives a flat, 23–31% higher integral
- ▶ the physical total is a one-bin extraction: full  $\cos\theta_\mu$  range,  $A_C \equiv 1$

## Total flux-averaged cross section ( $10^{-38} \text{ cm}^2/\text{Ar}$ )

|       | inclusive       | proton-tagged   |
|-------|-----------------|-----------------|
| FHC   | $1.03 \pm 0.40$ | $0.62 \pm 0.26$ |
| RHC   | $0.93 \pm 0.35$ | $0.53 \pm 0.25$ |
| Comb. | $0.97 \pm 0.37$ | $0.57 \pm 0.24$ |

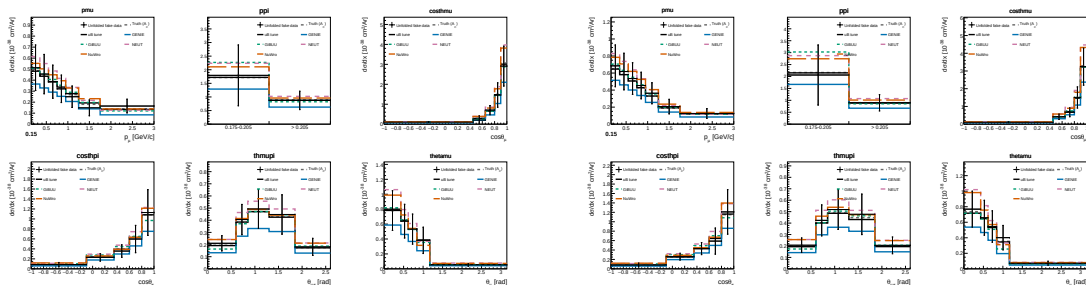
Extracted on the thrown fake data with the full universe covariance; closes on the realised truth to 0.975–0.998. RHC/FHC of the tune = 0.90. Generators span  $\times 1.6$ , all within  $1\sigma$  of the  $\sim 38\%$  band: no discrimination at this exposure.

# Differential results, FHC



Unfolded fake data,  $A_C$ -smeared truth, tune and the four generators, all in the  $A_C$ -smeared space. Generators differ by normalisation ( $\times 1.6$ , GENIE to NEUT); shapes agree within the uncertainties. Full covariance: flux, cross section, re-interaction, detector, POT, targets, statistics.

# RHC and combined

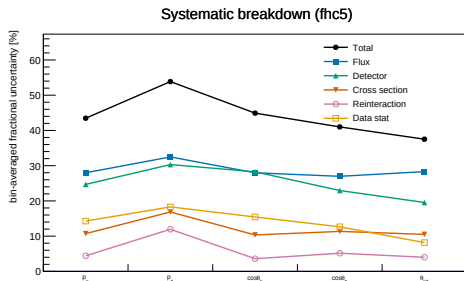


RHC

Combined FHC+RHC

Same 600 PPFX universes index-matched across both horn modes (rigorous flux correlation); native Run-4 RHC detector variations; each mode normalised to its own exposure and flux.

# Uncertainties



Bin-averaged fractional uncertainty, inclusive (%):

| Source          | range |                         |
|-----------------|-------|-------------------------|
| Flux (PPFX)     | 24–33 | dominant                |
| Detector        | 16–35 | ~ flux for several obs. |
| Cross section   | 9–17  |                         |
| Re-interaction  | 4–12  |                         |
| Data statistics | 6–18  |                         |
| Total           | 32–54 |                         |

- ▶ the  $\sim 17\%$  flux normalisation becomes  $\sim 30\%$  through the correlated background subtraction,  $17\% \times (1 + B/S)$ ; the one-bin total (30.2%, no regularisation) shows it exactly
- ▶ all 51 extractions carry a full covariance including detector variations;  $A_C$  and covariances released as text

# Validation, and what the review found

## Validation

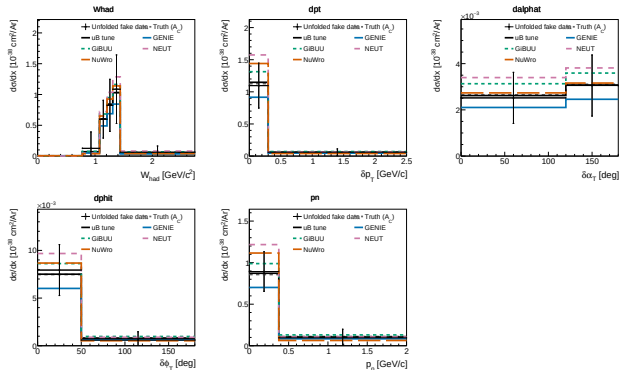
- ▶ fake-data closure: unfolded/realised truth = 1.010 on average over the 18 inclusive extractions; all  $p > 0.9$
- ▶ 32-throw ensemble (current release): pull widths 1.08 ( $p_\mu$ ), 0.97 ( $\cos\theta_\mu$ ), 1.02 (one-bin total); 62–71% / 92–94% coverage; means 0.5–2.7% low, at most  $1.9\sigma$ , stated not corrected
- ▶ clean-cache rebuild of all 51 extractions bit-identical (2026-09-02 pass; to be repeated before unblinding)
- ▶ five automated guards: normalisation manifest, systematics/file coupling, staleness, table consistency, blinding guard

## Corrected since August (all in the change log)

- ▶ beam-off scaled by an *event* count instead of gates: EXT was  $9.7\times$  too large ( $128 \rightarrow 13$  FHC events), then per run period ( $13 \rightarrow 30$ )
- ▶ fake-data truth carried the CV weight twice (1.6% on closure)
- ▶ PPFX CV weight missing from count-level tables (3–6%)
- ▶ one safe-weight rule everywhere; fake data re-thrown with it (1%)
- ▶  $A_C$  applied twice to models; TKI computed about detector  $z$  (both withdrawn and redone)

None of these touched the real-data path; several would have biased a real-data result.

# Proton-tagged subsample

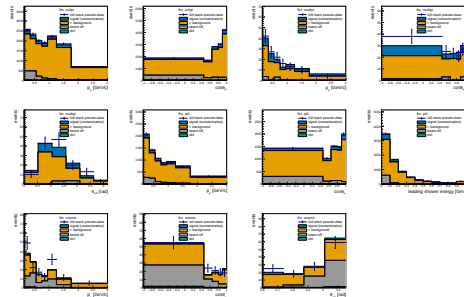


- ▶ inclusive selection + a contained proton  $> 0.3 \text{ GeV}/c$ ; efficiency 11.4%, purity  $\sim 50\%$ ; strong cosmic rejector
- ▶ TKI ( $\delta p_T$ ,  $\delta\alpha_T$ ,  $\delta\phi_T$ ,  $p_n$ ) in two bins, beam frame;  $W_{\text{had}}$  in six
- ▶ single-nucleon STV prescription applied to the  $\pi p$  system:  $p_n$  is a proxy (stated limitation)
- ▶ generators: normalisation spread only; no FSI discrimination at this exposure
- ▶ headline: one-bin total  $0.62 \pm 0.26$  (FHC)

# Control regions

| Region                  | Inversion                           | Role; $T = N^{\text{SR}}/N^{\text{CR}}$ of the target class                                                                                                  |
|-------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CC0 $\pi$               | $n_\pi = 0$                         | tests 0 $\pi$ feed-in; $T = 0.03$ , $\delta T/T = 31\text{--}34\%$ (detector-limited); target purity 72%                                                     |
| multi- $\pi$<br>$\pi^0$ | $n_\pi \geq 2$<br>shower veto fails | diagnostic only; $T = 2.5$<br>$\pi^0$ leakage; target CC+NC $\pi^0$ (ratified), purity 50–51%; $T = 0.02$ , $\delta T/T = 17\text{--}19\%$ (MC-stat limited) |
| cosmic                  | $\theta_{\mu\pi} > 2.6$ rad         | beam-off norm. and shape; 34–41% EXT purity, no fitted correction                                                                                            |

- ▶ disjoint from the signal region event by event; multi- $\pi$  and  $\pi^0$  overlap ( $\sim 40\%$  of multi- $\pi$ )
- ▶ pseudo-data/prediction 0.99–1.12 everywhere; full-stack test (MC+EXT+dirt) at unity within statistics
- ▶ per-region systematics incl. detector variations; transfer factors as correlated ratios (flux cancels to 2–3%)



Frozen distributions and binning, FHC, full-stack pseudo-data in place of beam-on points.

# Request to the collaboration

Open the four inclusive control regions; the signal region and the proton-tagged regions stay blind

Frozen before opening: selections at branch level, run-dependent normalisation, beam-off scaling, displayed observables and binning, covariance, goodness-of-fit and the permitted responses.

## Procedure (to be ratified)

- ▶ FHC and RHC separately, combined as cross-check; data vs. MC+EXT+dirt
- ▶ pass: global normalisation  $\chi^2$  over the eight (region, mode) totals,  $p > 0.05$ ; per-region pulls (full and stat+detector) and normalisation-marginalised shape tests as diagnostics
- ▶ allowed: no change; added uncertainty; one-parameter normalisation of the target class (purity  $> 50\%$ , LR test, transfer uncertainty added); else model revision and renewed approval
- ▶ forbidden: any signal cut or binning change; generator choice after inspection; bin-by-bin reweighting

## Prerequisites

- ✓ tables from one release, checked automatically
- ✓ definitions, overlap = 0, plots and binning frozen
- ✓ EXT gates released per run; pairings asserted
- ✓ real beam-on files refused without XSEC\_UNBLIND=1
- ✓ full-stack closure; region systematics incl. detector; transfer uncertainties
- ☐ control-region skim of the beam-on files (signal-region events never written)
- ☐ ratification of the procedure

Before the signal region: PPFX-corrected flux constants confirmed; discrepancy treatment; clean-cache reproduction.

# Summary

- ▶ A blind, fully validated extraction of  $\nu_\mu + \bar{\nu}_\mu$  CC1 $\pi^\pm$  cross sections on argon in the NuMI beam: total and six differential distributions in FHC, RHC and combined, with full covariances and released  $A_C$  matrices.
- ▶ Two physics limits set the scope: the pion range saturates (two-bin  $p_\pi$ ; no  $W_{\pi p}$  shape), and  $A_C$  is not norm preserving (the total is a one-bin extraction).
- ▶ Fake-data totals  $1.03/0.93/0.97 \times 10^{-38} \text{ cm}^2/\text{Ar}$  with 38–39% uncertainty, flux- and detector-limited.
- ▶ A sequence of normalisation and weighting defects was found by review and fixed; automated guards now stop each of them from recurring.
- ▶ We ask to open the four inclusive control regions under the frozen procedure.

Documents: analysis note (Draft 1.5), technical supplement, change log; data release in `report/data_release/`.

**Backup**

# Backup: exposure and normalisation

| Configuration | Runs                   | POT ( $10^{20}$ ) |
|---------------|------------------------|-------------------|
| FHC           | 1, 2, 4, 5             | 8.857             |
| RHC           | 1, 2, 3, 4a–c          | 11.082            |
| Combined      | all (14 overlay files) | 19.939            |

|       | FHC   | RHC   |
|-------|-------|-------|
| Run 1 | 3.283 | 0.605 |
| Run 2 | 1.268 | 2.591 |
| Run 3 | –     | 5.003 |
| Run 4 | 2.075 | 2.883 |
| Run 5 | 2.231 | –     |

Beam-on POT per run,  $10^{20}$ .

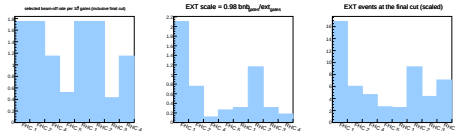
- ▶ every MC file scaled by  $P_{\text{data}}(\text{run})/P_{\text{MC}}(\text{file})$ ; multi-file runs normalised by the sum of distinct file POT
- ▶ combined configuration = per-mode data exposures, one universe chain (600 PPFx universes index-matched across FHC and RHC)
- ▶ integrated flux (new-G4,  $E_\nu > 60$  MeV): FHC 6.81, RHC 6.45, combined  $6.61 \times 10^{-10} \text{ cm}^{-2}/\text{POT}$  (exposure-weighted); PPFx-corrected constants to be confirmed before unblinding
- ▶  $N_{\text{Ar}} = 8.71 \times 10^{29}$  in the fiducial volume (dead  $z$  slab removed)
- ▶ CV weight = tune  $\times$  PPFx  $\times$  normalisation; safe rule: non-finite, negative or  $> 30 \rightarrow 1$ , zero kept. GENIE universes carry the tune; PPFx universes are absolute; re-interaction universes relative

# Backup: beam-off per run period

| Mode | Run | Beam-on gates | EXT files       | EXT gates  | scale |
|------|-----|---------------|-----------------|------------|-------|
| FHC  | 1   | 9 846 635     | run1            | 4 582 248  | 2.106 |
| FHC  | 2   | 3 535 129     | run1 (stand-in) | 4 582 248  | 0.756 |
| FHC  | 4   | 4 131 149     | run4a–d         | 34 831 149 | 0.116 |
| FHC  | 5   | 5 154 196     | run5            | 19 256 341 | 0.262 |
| RHC  | 1   | 1 458 253     | run1            | 4 582 248  | 0.312 |
| RHC  | 2   | 5 422 907     | run1 (stand-in) | 4 582 248  | 1.160 |
| RHC  | 3   | 10 349 610    | run3b           | 32 649 129 | 0.311 |
| RHC  | 4   | 6 304 167     | run4a–d         | 34 831 149 | 0.177 |

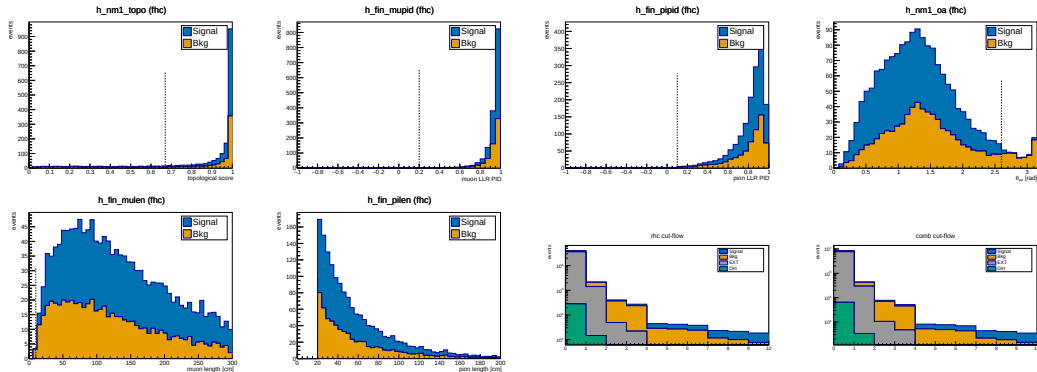
scale =  $0.98 \times \text{beam-on gates} / \sum \text{EXT gates}$ .

- ▶ cosmic-only data: the horn label is irrelevant, only the run period (CRT, detector conditions) matters
- ▶ no Run-2 beam-off exists; Run 1 is the stand-in (same pre-CRT conditions); Run 4 pools 4a–4d
- ▶ final-cut EXT: 30.2 (FHC), 23.2 (RHC), 53.4 (combined) events, 1.9% of the sample
- ▶ the previous release scaled by an *event* count:  $9.7\times$  too much EXT



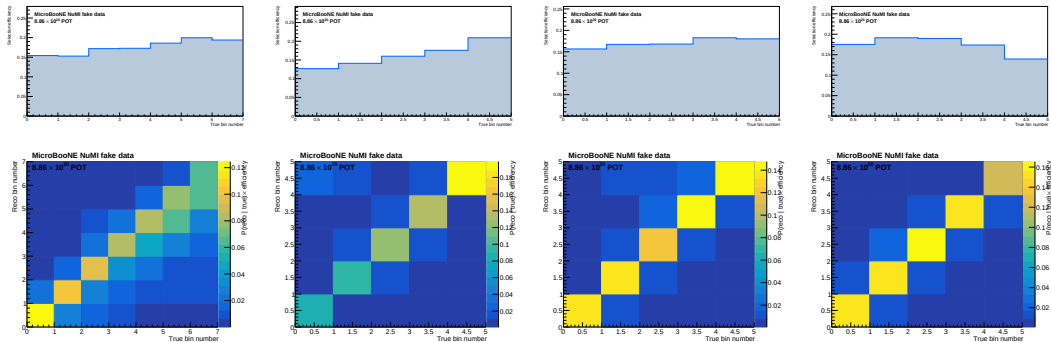
Selected cosmic rate per million gates: pre-CRT  
Run 1 is  $1.5\text{--}4\times$  the CRT-era runs (rates carry  
16–35% Poisson errors).

# Backup: selection variables (FHC)



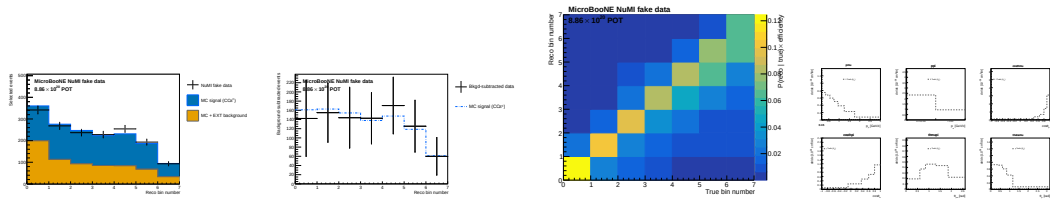
Top: topological score ( $N - 1$ ), muon-BDT score, pion-BDT score, opening angle ( $N - 1$ ). Bottom: muon and pion track lengths at the final cut; RHC and combined cut-flows. All per-run POT-scaled, full CV weight, fake data as points.

# Backup: efficiency and response



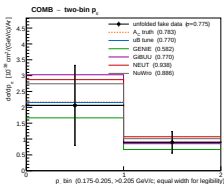
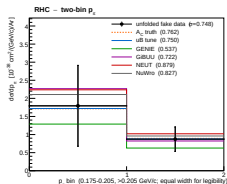
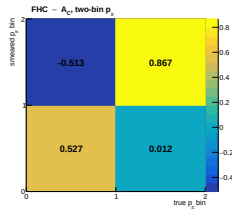
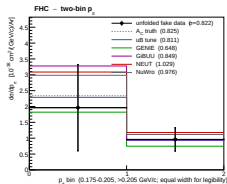
Top: selection efficiency vs. true variable (FHC). Bottom: smear acceptance  $S = \varepsilon M$  used in  $n^{\text{reco}} = S N^{\text{true}} + b$ ; top true bins are open-ended. Efficiency is folded into  $S$  once; no separate  $1/\varepsilon$  division.

# Backup: from reco distributions to the unfolded result ( $p_\mu$ , FHC)



- ▶ left to right: reco-level stack with fake data; background-subtracted; the smearacceptance  $S$ ; unfolded cross section with the  $A_C$ -smeared truth and models
- ▶ the Wiener-SVD regulariser is a second-derivative operator in bin index, corrected for the non-uniform bin widths (`fw_regmatrix`)
- ▶ the released products per extraction:  $A_C$ , unfolded values, full covariance, and the smeared predictions

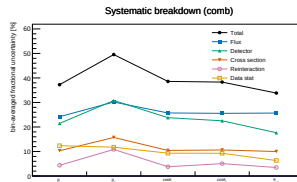
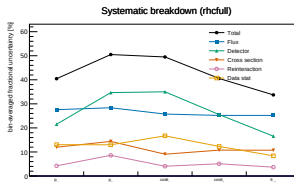
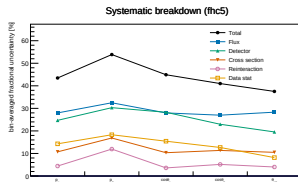
# Backup: two-bin pion momentum



Two-bin  $d\sigma/dp_\pi$  and its  $A_C$ : FHC (top), RHC and combined (bottom).

- ▶ range-based  $p_\pi$  under the muon hypothesis with the mass correction applied in the bin configuration; saturates at  $\sim 0.34$  GeV/c because most pions interact before stopping
- ▶ five-bin diagnostic: top true bin only 6% diagonal (criterion  $> 0.68$ )
- ▶ estimators tried and rejected: golden-pion tagging ( $40 \rightarrow 53\%$  golden at 87% efficiency, insufficient), MCS, calorimetric, regression on track variables, daughter recovery
- ▶ two bins  $[0.175, 0.205]$  and  $> 0.205$  GeV/c: diagonals 91% / 76%, released for all three configurations and the proton-tagged subsample
- ▶ same limit removes the proton-tagged  $W_{\pi p}$  shape (pion momentum enters directly)

# Backup: uncertainty breakdown by observable and configuration



Flux: 600 PPFX universes.

Cross section: GENIE multisim  
+ unisims.

Re-interaction: Geant4  
reweight.

Detector: Run-4 variation  
samples, FHC and RHC na-  
tively; proton-tagged included.  
POT 2%, targets 1%, statistics  
of MC and fake data.

Detector variations are 16–  
35% after unfolding (inclusive;  
10–109% proton-tagged).

Bin-averaged fractional uncertainty by source and observable, FHC, RHC and combined (2026-09-06 release).

# Backup: closure and ensemble tests

## Fake-data closure (18 inclusive extractions)

- ▶ fake data: an external Poisson throw of the CV prediction per run, weights reset to one in both reco and truth
- ▶ unfolded / realised truth = 1.010 on average;  $\chi^2$  probabilities  $> 0.9$  everywhere
- ▶ no alternative-generator fake-data test exists on the released chain yet (a NuWro/GiBUU pass is a prerequisite for unblinding)
- ▶ the  $A_C$ -smeared integral is observable-dependent by construction; D'Agostini gives a flat integral 23–31% higher, iteration-stable (it returns its prior on CV input)

## 32-throw ensemble

- ▶ pull widths 1.08 ( $p_\mu$ ), 0.97 ( $\cos\theta_\mu$ ) and 1.02 (one-bin total); 68% and 95% coverage recovered within the ensemble statistics
- ▶ ensemble means 0.5–2.7% below the reference (at most  $1.9\sigma$ ; largest in the most regularised observable), stated and not corrected
- ▶  $\cos\theta_\mu$ : the strongly suppressing  $A_C$  does not distort the uncertainties
- ▶  $\theta_\mu$  and  $\cos\theta_\mu$  are different partitions of one phase space; after the width-aware regularisation both discriminate,  $\theta_\mu$  with systematically larger  $\chi^2$  (model statements use  $\cos\theta_\mu$ )

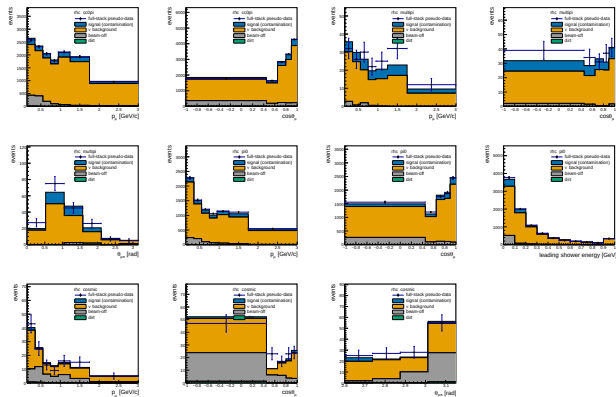
**Reproducibility:** clean-cache rebuild of all 51 extractions bit-identical on the 2026-09-02 pass, to be repeated on the release before unblinding; the universe cache is keyed on the normalisation scheme and on the file list.

# Backup: control-region transfer factors (FHC)

| Class                                       | SR    | CC0 $\pi$ | multi- $\pi$ | $\pi^0$ | cosmic | $T_{CC0\pi}$ | $T_{\text{multi}}$ | $T_{\pi^0}$ | $T_{\text{cosmic}}$ |
|---------------------------------------------|-------|-----------|--------------|---------|--------|--------------|--------------------|-------------|---------------------|
| signal (contamination)                      | 943   | 1 124     | 29.7         | 662     | 6.0    | 0.84         | 32                 | 1.43        | 158                 |
| $\nu_\mu$ CC 0 $\pi$ (track taken as pion)  | 244   | 8 781     | 2.6          | 1 011   | 23.2   | <b>0.03</b>  | 95                 | 0.24        | 11                  |
| $\nu_\mu$ CC 1 $\pi^\pm$ out of phase space | 52.7  | 629       | 2.7          | 361     | 14.8   | 0.08         | 19                 | 0.15        | 3.57                |
| $\nu_\mu$ CC with a $\pi^0$                 | 58.0  | 408       | 20.2         | 3 070   | 2.5    | 0.14         | 2.87               | <b>0.02</b> | 23                  |
| $\nu_\mu$ CC $\geq 2\pi^\pm$ , no $\pi^0$   | 145   | 222       | 57.8         | 658     | 3.9    | 0.65         | <b>2.51</b>        | 0.22        | 37                  |
| out of fiducial volume                      | 47.4  | 480       | 3.3          | 253     | 18.5   | 0.10         | 14                 | 0.19        | 2.56                |
| NC and $\nu_e$ CC                           | 89.5  | 549       | 7.1          | 632     | 6.4    | 0.16         | 13                 | 0.14        | 14                  |
| total $\nu$ -MC                             | 1 581 | 12 192    | 124          | 6 646   | 75.3   |              |                    |             |                     |
| CC+NC $\pi^0$ (ratified target)             | 64.5  | 444       | 21.8         | 3 331   | 3.2    | 0.15         | 2.96               | 0.019       | 20                  |

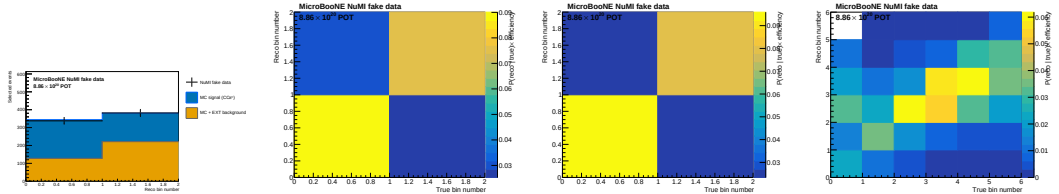
- ▶  $T = N^{\text{SR}}/N^{\text{CR}}$  per truth class; a normalisation change of the target class in a region propagates to the signal region by  $T$
- ▶ CC0 $\pi$ :  $T = 0.03$ ,  $\delta T/T = 31\text{--}34\%$  (genuine detector sensitivity);  $\pi^0$  (ratified target CC+NC $\pi^0$ , purity 50–51%):  $T = 0.02$ ,  $\delta T/T = 17\text{--}19\%$ , at its MC-statistical floor; flux cancels to 2–3% in the ratio
- ▶ the multi- $\pi$  region has  $T = 2.5$  and a 24% signal share: diagnostic only, no correction may be derived from it
- ▶ the cosmic region is 34–41% EXT with the rest dominated by CC0 $\pi$ ; it tests the beam-off normalisation and shape, no fitted correction

# Backup: control regions, RHC



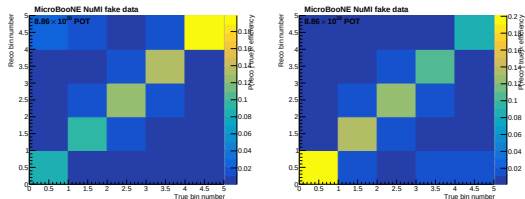
Frozen distributions ( $p_\mu$ ,  $\cos\theta_\mu$ , opening angle, shower energy) and binning for the four RHC regions; full-stack pseudo-data (MC+EXT+dirt, Poisson-thrown) in place of beam-on points. Pseudo-data/prediction 0.99–1.12.

# Backup: proton-tagged kinematics



- TKI built in the NuMI beam frame from the  $\mu$  and the leading proton with the pion added to the hadronic system; before the beam-frame fix  $\delta p_T$  was computed about detector  $z$  (0.74 vs. 0.26 GeV/c mean)
- two-bin releases for  $\delta p_T$ ,  $\delta \alpha_T$ ,  $\delta \phi_T$ ,  $p_n$ ; six bins in  $W_{\text{had}}$ ;  $W_{\pi p}$  withdrawn (pion momentum)
- all proton-tagged extractions carry detector variations (10–109%, median 33%; largest for six-bin  $W_{\text{had}}$ )
- the two-bin shapes separate generators by normalisation only; FSI discrimination would need the full exposure and better  $p_\pi$

# Backup: angles in the beam frame



Smearacceptance of  $\cos \theta_\mu$  (left) and  $\theta_\mu$  (right), beam frame, current release: two partitions of the same phase space (the  $\theta_\mu$  edges are not the arccos images of the  $\cos \theta_\mu$  edges).

- ▶ beam direction: truth uses the per-event neutrino direction; reco uses target-to-vertex, with the convention of the NuMI  $\nu_e$  analysis
- ▶ 64.6% of events changed  $\cos \theta_\mu$  bin when the axis was corrected from detector  $z$  to the beam
- ▶  $p_\mu$ ,  $p_\pi$ ,  $\theta_{\mu\pi}$  and the selection are frame-independent and were unaffected
- ▶  $\theta_{\mu\pi}$  coarsened  $7 \rightarrow 5$  bins to limit  $A_C$  smearing;  $\cos \theta_\pi$  re-binned to five forward-split bins ( $-1, -0.1, 0.35, 0.6, 0.8, 1$ ) after the 2026-09-05 re-check: better-conditioned  $A_C$ , same per-bin uncertainties